IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro

Project Documentation format

1. Introduction

- **Project Title:** IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro
- **Team Members**: Chintharala Sunil Kumar **(Project Lead & Cloud Architect)**
  Dungala Lova Raju **(Backend Developer & AI Engineer)**
  Kothapalli Prasanth **(Frontend Developer & UI/UX Designer)**
  Aladasu Yamuna **(Data Engineer & Model Analyst)**

2. Project Overview

- **Purpose:** To provide a seamless digital interface for non-technical users to interact with databases using natural language.

- **Features:**

  - **NLP Interface**: Converts plain English into SQL commands via Gemini 1.5 Flash.
  - **Secure Access**: Utilizes environment variables for API key masking.
  - **Live Data Preview**: Displays database records in interactive tables.
  - **Error Handling**: Implements Regex to clean AI responses for safe SQL execution.

3. Architecture

- **Frontend:**
  - Built using **Streamlit**, which allows for rapid deployment of a data-driven web interface.
  - The architecture is component-based, using a sidebar for multi-page navigation (Home, About, Query).

- **Backend:**
  - Built with **Python** and the **Google Generative AI SDK** to handle LLM processing.
  - Includes a logic layer that performs **Regex-based sanitization** to isolate raw SQL from AI responses.

- **Database:**
  - Uses **SQLite** for lightweight, relational data storage (replacing MongoDB).
  - The schema consists of a `STUDENTS` table with columns for **NAME**, **CLASS**, **SECTION**, **MARKS**, and **COMPANY**.

4. Setup Instructions

- **Prerequisites:**

    - Python 3.9+
    - Google Gemini API Key
    - Pip (Python package manager)

- **Installation:**

    1. **Clone the repository**: git clone [GitHub-URL]
    2. **Install dependencies**: pip install streamlit google-generativeai python-dotenv
    3. **Environment Variables**: Create a .env file in the root directory and add GOOGLE_API_KEY="your_api_key_here".

5. Folder Structure

- **Client:**
- **`app.py`:** Main entry point containing the Streamlit UI and AI integration logic.
- **`readme.md`**: Local documentation providing instructions on how to interact with the user interface.
- **`requirements.txt`**: List of dependencies required for the client-side libraries, such as `streamlit` and `google-generativeai`.

- **Server:**
- **`.env`**: The server configuration file that securely stores the `GOOGLE_API_KEY`.
- **`sql.py`**: The backend database engineering script used to initialize the server-side database and seed it with initial data.
- **`data.db`**: The SQLite database file where the server stores and retrieves all relational records.
- **`.gitignore`**: Server-side configuration to ensure sensitive files like `.env` are not exposed to version control.
- **pyvenv.cfg**: Server configuration file that manages the Python environment version and paths.

6. Running the Application

Step 1: Database Setup:

Bash

python sql.py

``` [cite: 148]

**Step 2: Start Application:**
```

Bash

```
streamlit run app.py
```
[cite: 146]

App URL: The dashboard is accessible at **http://localhost:8501**

7. API Documentation

SQL Generation:

- **Method**: Internal POST to Gemini 1.5 Flash.
- **Parameters**: prompt_context, user_question.
- **Response**: A SQL string (e.g., SELECT * FROM STUDENTS;).

Database Execution:

- **Function**: read_query(sql, db).
- **Parameters**: Cleaned SQL string and database path.

8. Authentication

- **API Authentication**: Handled via **API Key-based authentication** using the Google Generative AI SDK.
- **Security**: Keys are stored in an encrypted/masked .env file and loaded into the environment at runtime using python-dotenv.

9. User Interface

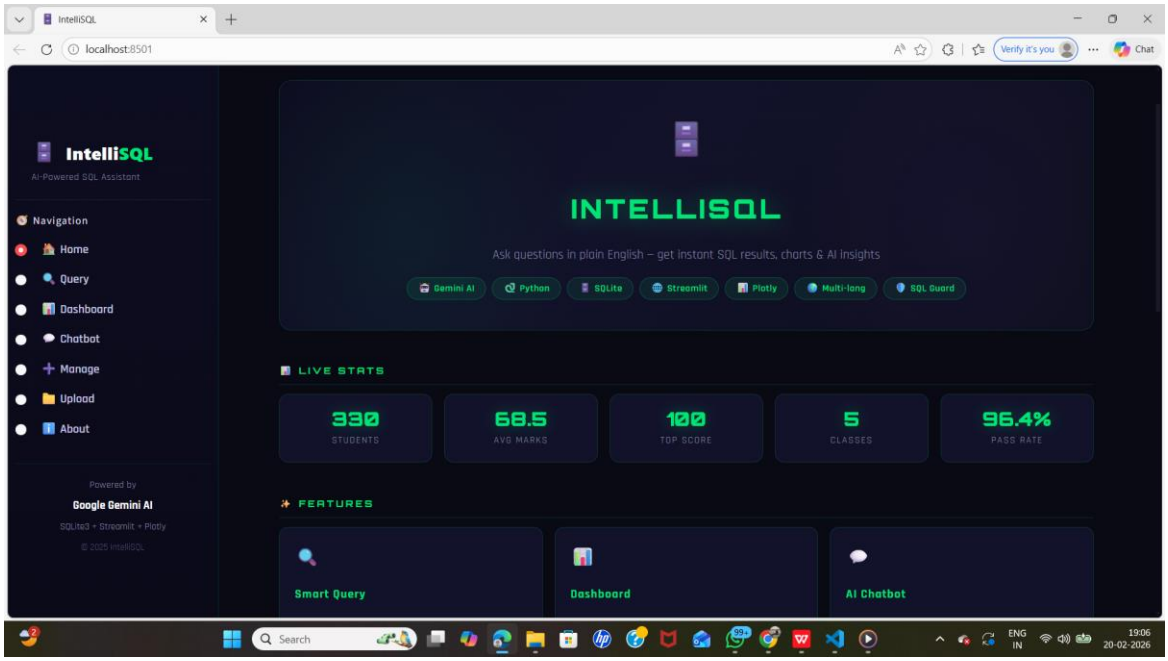
- **Registration/Login**: Simplified via API key validation upon app startup.
- **Query Interface**: A professional text-input area where users type natural language questions.
- **Data Visualization**: Real-time rendering of results in interactive data tables.

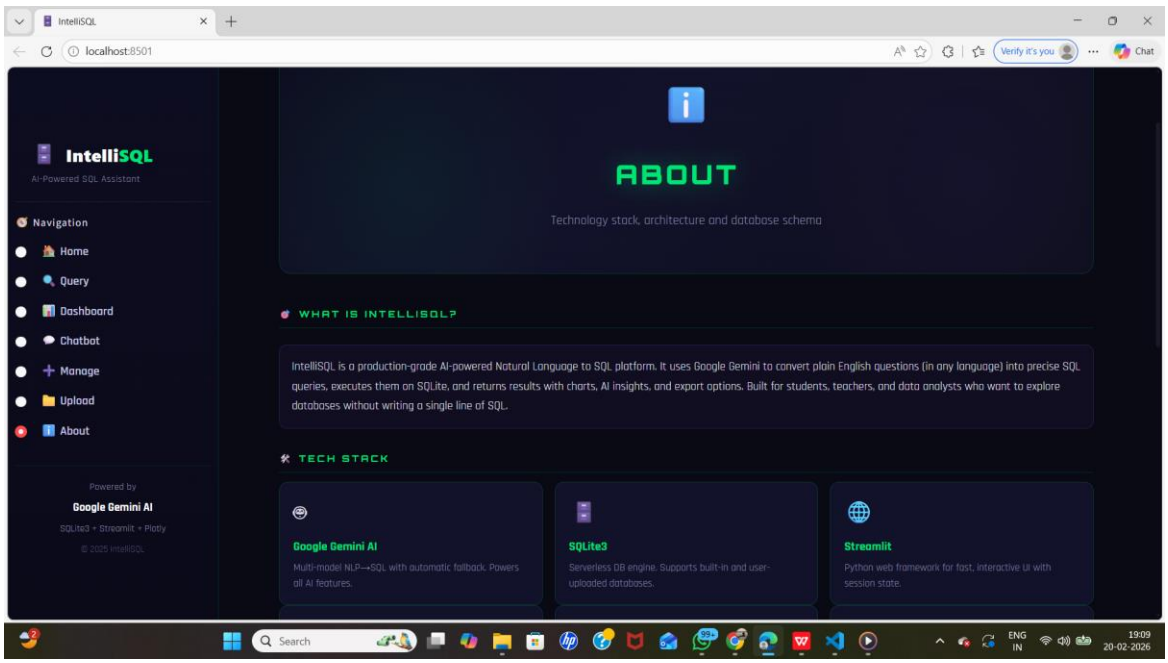
10. Testing

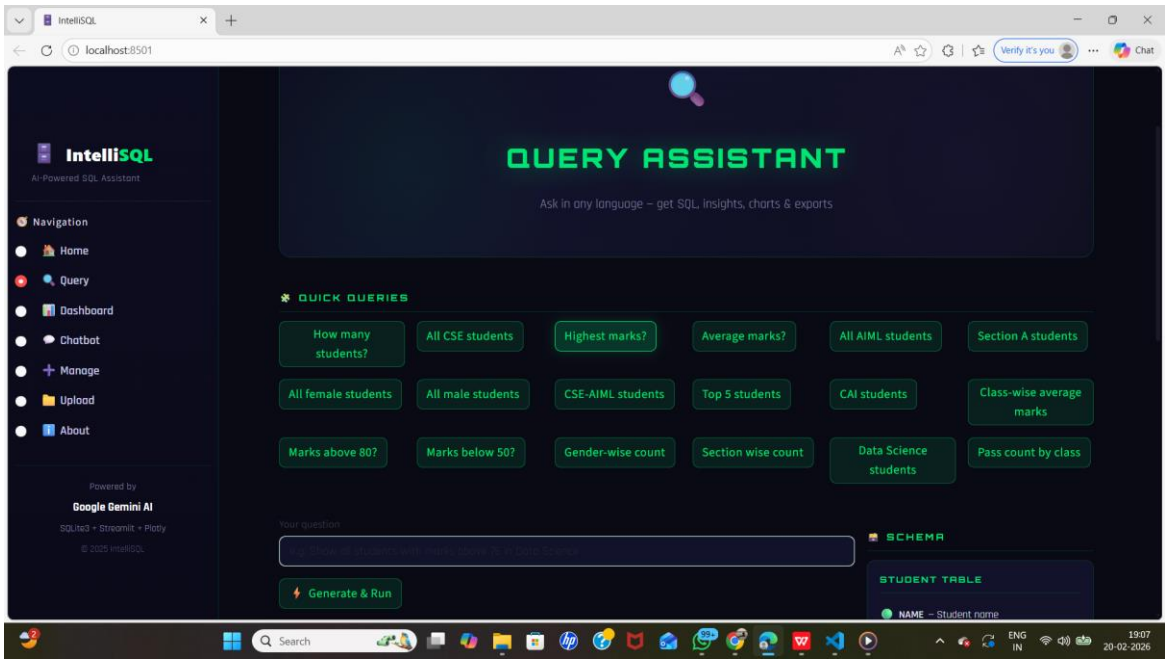
- **Strategy**: Functional testing was performed on diverse natural language phrasings to ensure SQL accuracy.
- **Tools**: **Manual UAT** and **Python Debugger** were used to verify that Regex correctly strips AI conversational text.

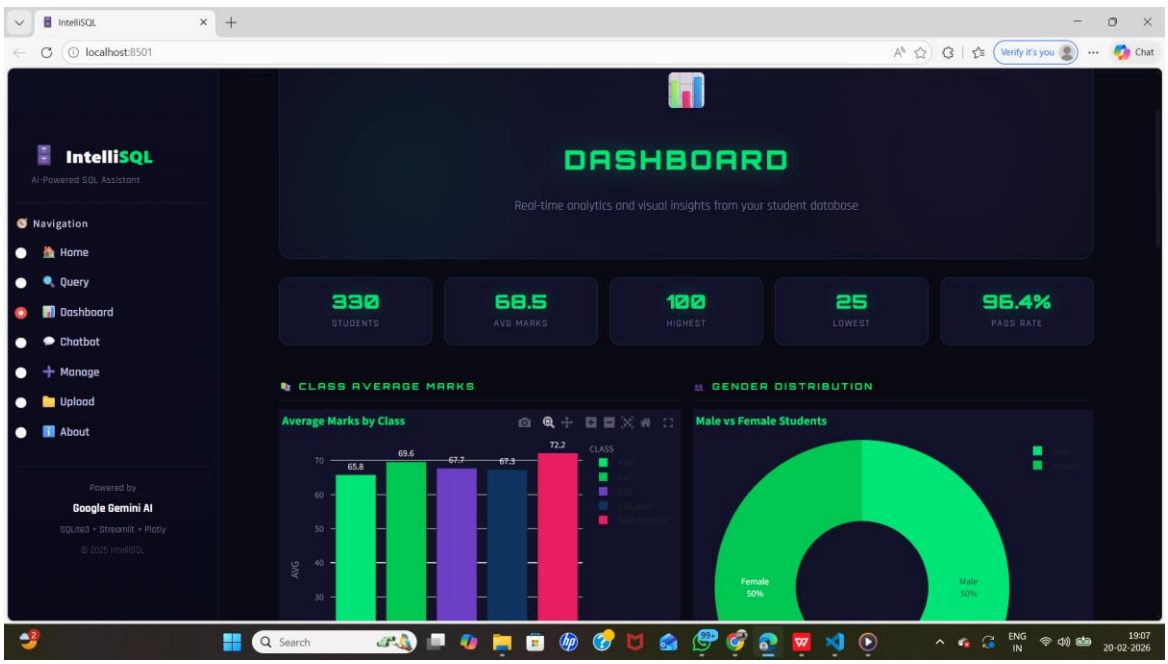
11. Screenshots of project

GitHub Repository:






IntelliSQL
localhost:8501
IntelliSQL
AI-Powered SQL Assistant
Navigation
Home
Query
Dashboard
Chatbot
Manage
Upload
About
Powered by
Google Gemini AI
SQLite3 + Streamlit + Plotly
© 2025 IntelliSQL
QUERY ASSISTANT
Ask in any language – get SQL, insights, charts & exports
QUICK QUERIES
How many students?
All CSE students
Highest marks?
Average marks?
All AIML students
Section A students
All female students
All male students
CSE-AIML students
Top 5 students
CAI students
Class-wise average marks
Marks above 80?
Marks below 50?
Gender-wise count
Section wise count
Data Science students
Pass count by class
Your question
Generate & Run
SCHEMA
STUDENT TABLE
NAME – Student name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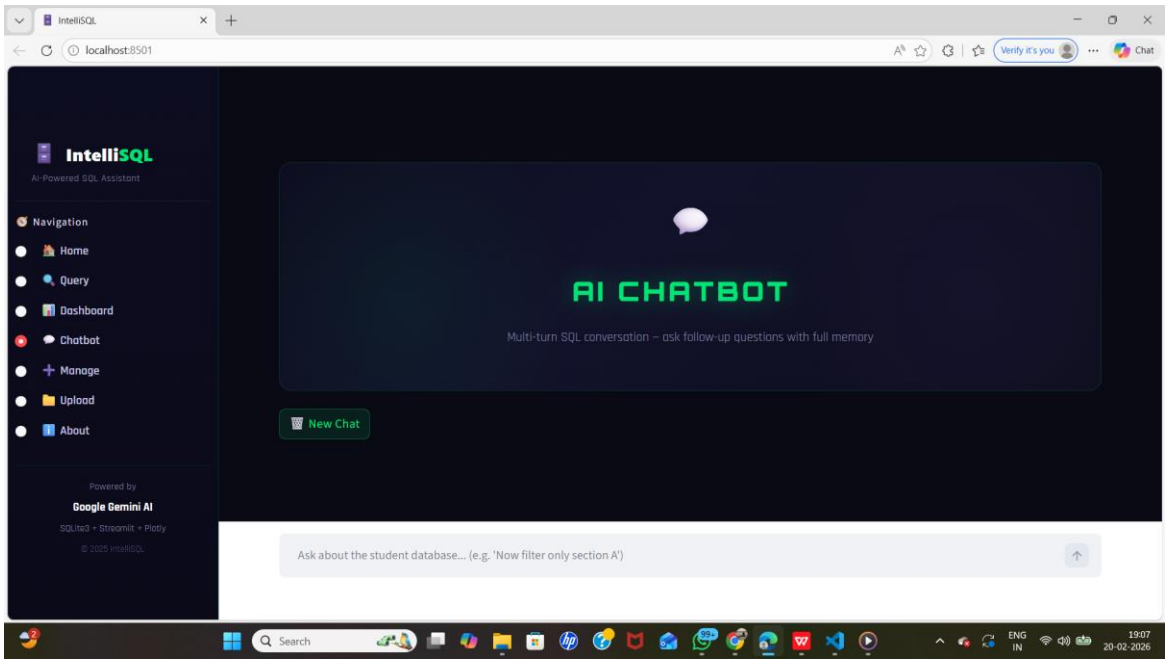
DASHBOARD
Real-time analytics and visual insights from your student database
330
STUDENTS
68.5
AVG MARKS
100
HIGHEST
25
LOWEST
96.4%
PASS RATE
CLASS AVERAGE MARKS
Average Marks by Class
65.8
69.6
67.7
67.3
72.2
CLASS
AVG
GENDER DISTRIBUTION
Male vs Female Students
Female 50%
Male 50%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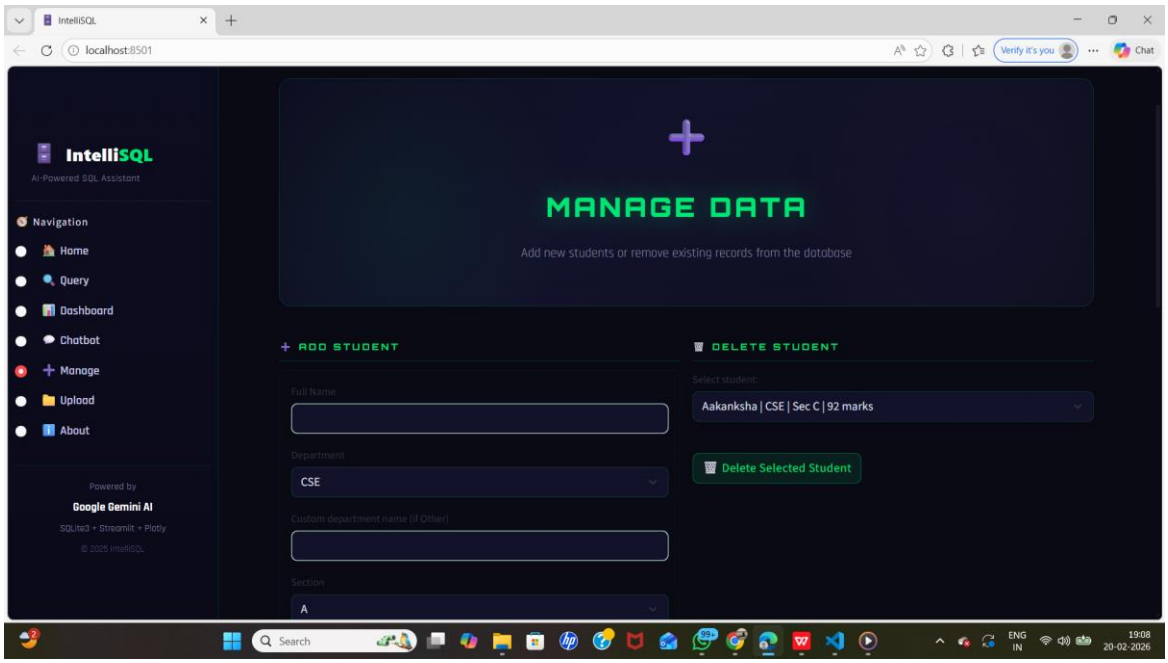
AI CHATBOT
Multi-turn SQL conversation – ask follow-up questions with full memory
New Chat
Ask about the student database... (e.g. 'Now filter only section A')
Search
ENG
IN
19:07
20-02-2026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Query
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Manage
Upload
About
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Google Gemini AI
SQLite3 + Streamlit + Plotly
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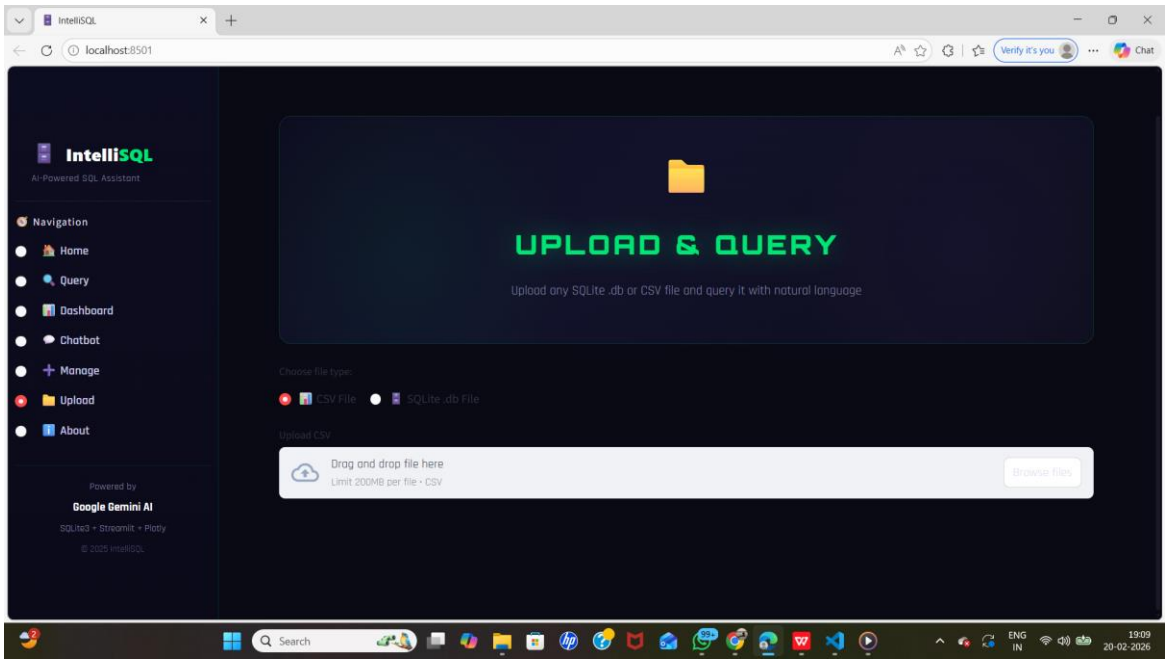
MANAGE DATA
Add new students or remove existing records from the database
ADD STUDENT
Full Name
Department
CSE
Custom department name (if Other)
Section
A
DELETE STUDENT
Select student:
Aakanksha | CSE | Sec C | 92 marks
Delete Selected Student
Search
ENG
IN
19:08
20-02-2026



12. Known Issues

- **Complex Joins**: The current model may struggle with multi-table queries if the schema isn't explicitly defined in the prompt.
- **Responsiveness**: Wide data tables may require horizontal scrolling on smaller mobile displays.

13. Future Enhancements

- **Voice-to-SQL**: Integration of a microphone icon to allow users to speak their queries.
- **AI Insights**: Using the AI to explain the data results in plain English below the table.
- **Multi-DB Support**: Adding connectors for PostgreSQL and MySQL.